

① Determine whether the following integral is convergent or divergent

$$\int_1^{\infty} \frac{1 + \sin \sqrt{x}}{\sqrt{x} + x^3} dx = ?$$

Solution: $\int_1^{\infty} \frac{2}{x^3} dx$ is convergent by p-test

$$\frac{1 + \sin \sqrt{x}}{\sqrt{x} + x^3} \leq \frac{2}{\sqrt{x} + x^3} \leq \frac{2}{x^3}, \quad \text{so by comparison}$$

$$\int_1^{\infty} \frac{1 + \sin \sqrt{x}}{\sqrt{x} + x^3} dx \quad \text{is convergent.}$$

② Evaluate the integral $\int_1^{\infty} \frac{\ln x}{x^3} dx$, if it is convergent

$$\frac{\ln x}{x^3} < \frac{x}{x^3} = \frac{1}{x^2} \quad ; \quad \int_1^{\infty} \frac{1}{x^2} dx \quad \text{convergent, so}$$

$$\int_1^{\infty} \frac{\ln x}{x^3} dx \quad \text{is convergent.}$$

$$\int_1^{\infty} \frac{\ln x}{x^3} dx = \lim_{t \rightarrow \infty} \left[\ln x \cdot \left(-\frac{1}{2x^2}\right) \Big|_1^t + \int_1^t \frac{dx}{2x^3} \right]$$

$$\begin{aligned} \left. \begin{array}{l} \ln x = u \Rightarrow \frac{dx}{x} = du \\ \frac{dx}{x^3} = dv \Rightarrow -\frac{1}{2x^2} = v \end{array} \right\} &= \lim_{t \rightarrow \infty} \left[\left(\ln t \cdot \left(-\frac{1}{2t^2}\right) - \ln 1 \cdot \left(-\frac{1}{2 \cdot 1}\right) \right) - \left(\frac{1}{4t^2} \right) + \left[\frac{1}{4} \right] \right] \\ &= \frac{1}{4} \end{aligned}$$

$$\lim_{t \rightarrow \infty} \left(-\frac{\ln t}{2t^2} \right) \rightarrow 0$$

App. L'Hôpital's rule

③ Is the following integral convergent or divergent?
(Note you are not asked to evaluate the integral)

$$\int_8^{\infty} \frac{x+1}{\sqrt{x^4-x}} dx$$

$$x > 0; \quad \frac{x+1}{\sqrt{x^4-x}} > \frac{x}{\sqrt{x^4-x}} > \frac{x}{\sqrt{x^4}} = \frac{x}{x^2} = \frac{1}{x}$$

Now, $\int_8^{\infty} \frac{1}{x} dx$ diverges, So by comparison

$$\int_8^{\infty} \frac{x+1}{\sqrt{x^4-x}} dx \quad \underline{\text{diverges}}.$$

④ Is the following integral convergent or divergent?

$$\int_0^{\pi/2} \frac{\sin x}{x^{3/2}} dx$$

$f(x) = \frac{\sin x}{x^{3/2}}$ is not defined and continuous at $x=0$.

for $x \in (0, \pi/2]$, $g(x) = \sin x$

$$\sin x < x \Rightarrow 0 < \frac{\sin x}{x^{3/2}} < \frac{x}{x^{3/2}} = \frac{1}{x^{1/2}}$$

$$\int_0^{\pi/2} \frac{dx}{x^{1/2}} \text{ converges, because } p = \frac{1}{2} < 1.$$

So by comparison,

$$\int_0^{\pi/2} \frac{\sin x}{x^{3/2}} dx \quad \underline{\text{converges}}.$$

⑤ Homework: Evaluate $\int_0^2 \frac{\ln x}{\sqrt{2x}} dx$.

Answer: $(2\ln 2 - 4)$

⑥ Homework: Evaluate $\int_0^{\pi} \frac{dx}{1 + \cos x}$.

Answer: (Diverges)

⑦ Homework: Evaluate $\int_1^{\infty} \frac{dx}{\sqrt{x^2 - 0.1}}$.

Answer: (Diverges)

⑧ Determine whether the following integral is convergent $\int_1^{\infty} \frac{\sin^2 x}{x^2} dx$

9) $\int_1^{\infty} \frac{x-1}{x^2+5} dx = ?$

$\int_1^{\infty} \frac{dx}{x}$ diverges ($p=1$) and

$$\lim_{x \rightarrow \infty} \frac{\frac{x-1}{x^2+5}}{\frac{1}{x}} = 1.$$

Thus,

$\int_1^{\infty} \frac{x-1}{x^2+5} dx$ diverges because $0 < 1 < \infty$ and $\int_1^{\infty} \frac{dx}{x}$ diverges.

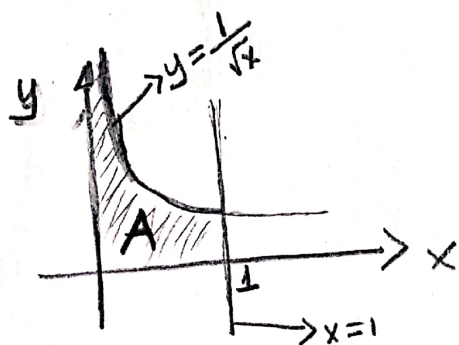
10) $\int_1^{\infty} \frac{x^2+7x}{x^5+x^4-2x+5} dx = ?$

$\int_1^{\infty} \frac{dx}{x^3}$ converges ($p=3$).

$$\lim_{x \rightarrow \infty} \frac{\frac{x^2+7x}{x^5+x^4-2x+5}}{\frac{1}{x^3}} = 1. \text{ Thus,}$$

$\int_1^{\infty} \frac{x^2+7x}{x^5+x^4-2x+5} dx$ converges because $0 < 1 < \infty$

and $\int_1^{\infty} \frac{dx}{x^3}$ converges.



Find the shaded area?

Answer: $A = 2$